

DBMS – PRODUCT AND CATEGORY MANAGEMENT

Project Title : Lenskart – Sustainable Eyewear E-Commerce Order Management System

Skill Task : DBMS

1. Database Design

1.1 Category Table

Attribute	Key Type
Category_ID	Primary Key
Category Name	Unique Key
Description	-

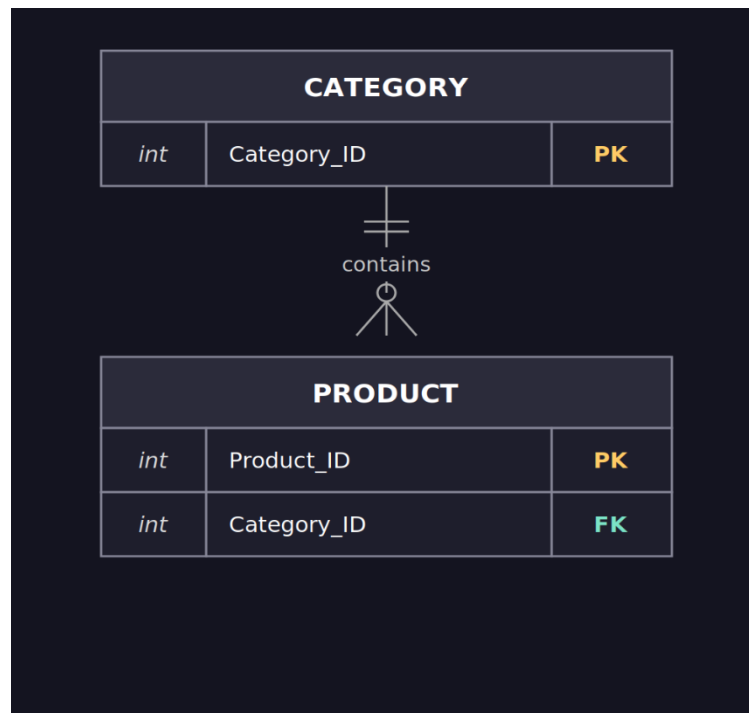
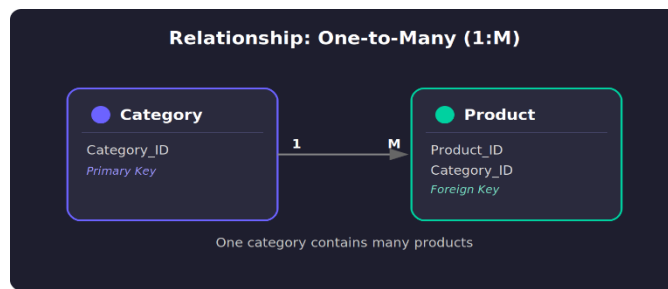
1.2 Product Table

Attribute	Key Type
Product_ID	Primary Key
Product Name	-
Category_ID	Foreign Key
Brand_ID	Foreign Key
Frame_Type	-
Price	-
Stock	-
Power_Compatible	-

2. Primary and Foreign Key Relationship

Category ----> Product

- Category.Category_ID ----> Primary Key
- Product.Product_ID ---- > Primary Key
- Product.Category_ID ---- > Foreign Key referencing Category.Category_ID
- Relationship : One-to-Many (1:M)



3. Table Creation

3.1 Create Category Table

```
SQL> CREATE TABLE Category (  
2     Category_ID NUMBER PRIMARY KEY,  
3     Category_Name VARCHAR2(100) UNIQUE,  
4     Description VARCHAR2(255)  
5 );
```

Table created.

3.2 Create Product Table

```
SQL> CREATE TABLE Product (  
 2 Product_ID INT PRIMARY KEY,  
 3 Product_Name VARCHAR2(100),  
 4 Category_ID INT,  
 5 Brand_ID INT,  
 6 Frame_Type VARCHAR2(50),  
 7 Price NUMBER(10,2),  
 8 Stock INT,  
 9 Power-Compatible VARCHAR2(10),  
10 FOREIGN KEY (Category_ID) REFERENCES Category(Category_ID)  
  
11 );  
  
Table created.
```

4. Data Insertion

4.1 Category Records

```
SQL> INSERT INTO Category  
 2 VALUES (1, 'Eyeglasses', 'Prescription and non-prescription eyeglasses');  
  
1 row created.  
  
SQL> INSERT INTO Category  
 2 VALUES (2, 'Sunglasses', 'Stylish and protective sunglasses');  
  
1 row created.  
  
SQL> INSERT INTO Category  
 2 VALUES (3, 'Computer Glasses', 'Blue light blocking glasses for screen use');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Category  
 2 VALUES (4, 'Kids Glasses', 'Eyewear specially designed for children');  
  
1 row created.  
  
SQL> INSERT INTO Category  
 2 VALUES (5, 'Eco-Friendly Frames', 'Sustainable eyewear made from recycled and natural materials');  
  
1 row created.
```

4.2 Product Records

```
SQL> INSERT INTO Product
  2 VALUES (101, 'Eco Green Round Glasses', 1, 201, 'Round', 1499.00, 50, 'Yes');

1 row created.

SQL> INSERT INTO Product
  2 VALUES (102, 'Bamboo Frame Spectacles', 2, 202, 'Rectangle', 1999.00, 35, 'Yes');

1 row created.

SQL> INSERT INTO Product
  2 VALUES (103, 'Recycled Plastic Sunglasses', 3, 203, 'Wayfarer', 1299.00, 40, 'No');

1 row created.

SQL> INSERT INTO Product
  2 VALUES (104, 'Wooden Finish Eyeglasses', 1, 204, 'Square', 2499.00, 25, 'Yes');

1 row created.

SQL> INSERT INTO Product
  2 VALUES (105, 'Eco-Friendly Kids Glasses', 4, 205, 'Oval', 999.00, 60, 'Yes');

1 row created.
```

5. Data Retrieval

5.1 Display Category Records

```
SQL> select * from category;

CATEGORY_ID
-----
CATEGORY_NAME
-----
DESCRIPTION
-----
          1
Eyeglasses
Prescription and non-prescription eyeglasses

          2
Sunglasses
Stylish and protective sunglasses

CATEGORY_ID
-----
CATEGORY_NAME
-----
DESCRIPTION
-----

          3
Computer Glasses
Blue light blocking glasses for screen use
```

```

      4
Kids Glasses
CATEGORY_ID
-----
CATEGORY_NAME
-----
DESCRIPTION
-----
Eyewear specially designed for children

      5
Eco-Friendly Frames
Sustainable eyewear made from recycled and natural materials

SQL> |

```

5.2 Display product Records

```

SQL> select * from product;

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  FRAME_TYPE
-----
      PRICE      STOCK  POWER_COMP
-----
      101
Eco Green Round Glasses
      1      201 Round
      1499      50 Yes

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  FRAME_TYPE
-----
      PRICE      STOCK  POWER_COMP
-----
      102
Bamboo Frame Spectacles
      2      202 Rectangle
      1999      35 Yes

```

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  FRAME_TYPE
-----
PRICE        STOCK    POWER_COMP
-----
103
Recycled Plastic Sunglasses
3           203 Wayfarer
1299        40 No

```

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  FRAME_TYPE
-----
PRICE        STOCK    POWER_COMP
-----
104
Wooden Finish Eyeglasses
1           204 Square
2499        25 Yes

```

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  FRAME_TYPE
-----
PRICE        STOCK    POWER_COMP
-----
105
Eco-Friendly Kids Glasses
4           205 Oval
999         60 Yes

```

SQL> |

6. Product Update

Update the Price and product Id of Eco Green Round Glasses.

```

SQL> UPDATE Product
2  SET Price = 1799.00
3  WHERE Product_ID = 101;

1 row updated.

```

7. Product Deletion

```

SQL> DELETE FROM Product
2  WHERE Product_ID = 104;

1 row deleted.

```

8. Category-wise Product Reports

8.1 Product Listing by Category

For your project report, you can write:

Product Listing by Category Report

This report displays products category-wise along with their Product ID, Product Name, Price, and Stock details. It helps in organizing and managing products efficiently.

```
SQL> SELECT p.Product_ID,  
2         p.Product_Name,  
3         c.Category_Name,  
4         p.Price,  
5         p.Stock  
6 FROM Product p  
7 JOIN Category c  
8 ON p.Category_ID = c.Category_ID  
9 ORDER BY c.Category_Name;
```

PRODUCT_ID

PRODUCT_NAME

CATEGORY_NAME

PRICE

STOCK

103

Recycled Plastic Sunglasses
Computer Glasses

1299

40

PRODUCT_ID

PRODUCT_NAME

CATEGORY_NAME

PRODUCT_ID	1799	Eyeglasses	50
PRODUCT_NAME			
CATEGORY_NAME			
PRICE	105		
STOCK	999	Eco-Friendly Kids Glasses	60
PRODUCT_ID			
PRODUCT_NAME			
CATEGORY_NAME			
PRICE	102		
STOCK	1999	Bamboo Frame Spectacles	35

8.2 Category-Based Product Listing

```
SQL> SELECT p.Product_ID,
2      p.Product_Name,
3      c.Category_Name
4 FROM Product p
5 JOIN Category c
6 ON p.Category_ID = c.Category_ID;
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_NAME
-----
101
Eco Green Round Glasses
Eyeglasses
```

```
104
Wooden Finish Eyeglasses
Eyeglasses
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_NAME
-----
102
Bamboo Frame Spectacles
```

```

102
Bamboo Frame Spectacles
Sunglasses

103
Recycled Plastic Sunglasses

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_NAME
-----
Computer Glasses

105
Eco-Friendly Kids Glasses
Kids Glasses

SQL> |

```

8.3 Product count by category

```

SQL> SELECT c.Category_Name,
2         COUNT(p.Product_ID) AS Product_Count
3 FROM Category c
4 LEFT JOIN Product p
5 ON c.Category_ID = p.Category_ID
6 GROUP BY c.Category_Name
7 ORDER BY Product_Count DESC;

```

```

CATEGORY_NAME
-----

```

```

PRODUCT_COUNT
-----

```

```

Eyeglasses      2

```

```

Sunglasses      1

```

```

Computer Glasses 1

```

```

CATEGORY_NAME
-----

```

```

PRODUCT_COUNT
-----

```

```

Kids Glasses     1

```

```

Eco-Friendly Frames 0

```